30-Day Beginner Lecture

Course Title

GPT + AI Agents + RAG + n8n + API Integration + Business Automation

Target Audience

Beginners who want to understand modern AI and learn how GPT, AI Agents, RAG, APIs, n8n, and automation work together in real business systems.

Course Duration

30 Days

60 Minutes per Day

Total: 30 Hours

Main Course Goal

By the end of the 30 days, students should understand how to:

- Use GPT effectively
- Understand LLM fundamentals
- Design good AI prompts
- Use structured AI outputs
- Understand AI Agents
- Connect agents with tools
- Understand RAG
- Build simple knowledge assistants
- Understand APIs and webhooks
- Build workflows with n8n
- Connect GPT to business applications
- Automate common business processes
- Design complete AI automation systems
- Build a beginner portfolio project

Standard 60-Minute Class Format

Use approximately this structure every day:

- **0–10 min:** Review + introduction
- **10–30 min:** Core concept
- **30–45 min:** Live demonstration
- **45–55 min:** Student exercise
- **55–60 min:** Q&A + homework

WEEK 1 — AI & GPT FUNDAMENTALS

Day 1 — Introduction to Generative AI

Goal

Understand what modern Generative AI is and why businesses are adopting it.

Topics

- What is Artificial Intelligence?
- Traditional AI vs Generative AI
- What is an LLM?
- What is GPT?
- Tokens
- Context
- Training vs inference
- AI capabilities
- AI limitations
- Hallucinations

Demonstration

Compare a normal Google-style search with a GPT conversation.

Exercise

Students ask GPT to:

1. Summarize text
2. Rewrite an email
3. Extract information
4. Generate ideas
5. Classify customer feedback

Homework

Find five business tasks where AI could save time.

Day 2 — Understanding GPT and LLMs

Goal

Understand how GPT-style models work conceptually.

Topics

- Large Language Models
- Tokens
- Token prediction
- Context windows
- Temperature
- Reasoning
- Model input
- Model output
- Why AI sometimes gives different answers
- Why AI makes mistakes

Demo

Input:

> Categorize this support request.

Output:

```
{
  "category": "refund",
  "priority": "high"
}
```

Exercise

Test the same question with different instructions.

Homework

Write three examples where GPT could replace repetitive manual work.

Day 3 — Prompt Engineering Fundamentals

Goal

Learn how to communicate effectively with AI.

Topics

A good prompt contains:

```
Role
+
Goal
+
Context
+
Instructions
+
Constraints
+
Output Format
```

Example

Bad:

```
Write marketing text.
```

Better:

```
You are an ecommerce marketing specialist.

Write a product description for a men's hair product.

Audience:
Men aged 18–35.

Tone:
Professional and energetic.

Length:
Maximum 150 words.

Return:
```

```
Headline
Description
3 benefits
CTA
```

Exercise

Students improve five bad prompts.

Homework

Create one reusable prompt for a business task.

Day 4 — Advanced Prompt Techniques for Beginners

Topics

- Zero-shot prompting
- Few-shot prompting
- Examples
- Constraints
- Roles
- Output formatting
- Prompt templates
- Prompt variables
- Asking AI to verify outputs

Demo

Create a reusable:

```
Customer Email Analyzer
```

Input:

```
Customer message
```

Output:

```
{
  "sentiment": "",
  "category": "",
```

```
  "priority": "",
  "summary": "",
  "recommended_action": ""
}
```

Exercise

Build an email classification prompt.

Day 5 — Structured AI Output

Goal

Understand why applications need structured responses.

Topics

Normal response:

```
This customer seems unhappy and probably wants a refund.
```

Structured response:

```
{
  "sentiment": "negative",
  "intent": "refund",
  "priority": "high"
}
```

Explain:

- JSON
- Fields
- Values
- Schemas
- Validation
- Why automation requires structured output

Exercise

Convert five natural-language AI responses into JSON.

Day 6 — AI Use Cases in Business

Topics

AI for:

Customer Support

- Ticket classification
- Reply drafting
- FAQ answering
- Sentiment analysis

Sales

- Lead qualification
- Personalized outreach
- Meeting summaries

Marketing

- Content generation
- Social posts
- SEO
- Product descriptions

Operations

- Document extraction
- Inventory analysis
- Reporting

HR

- Resume analysis
- Job descriptions
- Employee knowledge assistant

Exercise

Students choose one business and identify 10 AI opportunities.

Day 7 — Week 1 Mini Project

Project: AI Customer Support Assistant

Build a prompt that receives:

```
Customer message
```

and produces:

```
{
  "summary": "",
  "category": "",
  "sentiment": "",
  "priority": "",
  "recommended_response": "",
  "requires_human": false
}
```

60-Minute Class

10 min: Review
15 min: Project design
20 min: Build
10 min: Test
5 min: Discussion

WEEK 2 — AI AGENTS

Day 8 — What Is an AI Agent?

Goal

Understand the difference between ChatGPT and an agent.

Chatbot:

```
Question
↓
AI
```

```
↓
Answer
```

Agent:

```
Goal
↓
AI
↓
Decision
↓
Tool
↓
Result
↓
Next Decision
↓
Final Answer
```

Topics

- Agent
- Goal
- Tools
- Instructions
- Memory
- Environment
- Actions

Example

Customer:

```
Where is my order?
```

Agent:

```
Understand request
↓
Find order number
↓
Call order API
↓
Get shipping status
```

```
↓
Generate response
```

Day 9 — AI Agent Tools

Topics

Agents can use tools such as:

```
Search Knowledge
Get Customer
Check Order
Send Email
Create Ticket
Update CRM
Search Database
Book Meeting
```

Exercise

Design tools for:

AI Sales Agent

Example:

```
search_lead()
get_customer()
update_crm()
send_email()
schedule_meeting()
```

Day 10 — Function Calling

Goal

Understand how AI chooses functions.

Example tool:

```
get_weather(city)
```

User:

```
What is the weather in Denver?
```

AI decides:

```
Call get_weather("Denver")
```

Application executes it.

Result returns to AI.

AI generates the answer.

Business Example

```
get_order(order_id)
```

Exercise

Design five functions for a customer-support agent.

Day 11 — Agent Memory and Context

Topics

- Conversation history
- Short-term memory
- Long-term memory
- User preferences
- Database memory
- Session state
- Why unlimited memory is dangerous
- Privacy considerations

Demo

Conversation:

```
User: My company is ABC Fashion.

Later:

User: Create a marketing campaign for my company.
```

Discuss how context is retained.

Day 12 — Agent Decision Making

Topics

Agent loop:

```
Observe
↓
Think
↓
Choose Action
↓
Execute Tool
↓
Observe Result
↓
Continue / Stop
```

Discuss:

- tool selection
- failure handling
- confidence
- approval
- stopping conditions

Exercise

Design decision logic:

```
Refund request
↓
Order under $50?
↓
```

```
Yes → automated workflow
No → human approval
```

Day 13 — Single Agent vs Multi-Agent

Single Agent

```
Business Agent
├── Search
├── Email
├── CRM
└── Database
```

Multi-Agent

```
Manager Agent
│
├── Sales Agent
├── Support Agent
├── Finance Agent
└── Marketing Agent
```

Discussion

Explain why beginners should start with a single agent.

Exercise

Design a four-agent ecommerce system.

Day 14 — AI Agent Mini Project

Project: AI Sales Assistant

Architecture:

```
Lead
↓
AI Agent
```

```
├── Analyze Lead
├── Score Lead
├── Recommend Action
├── Generate Email
└── Recommend Follow-Up
```

Output:

```
{
  "lead_score": 85,
  "priority": "high",
  "recommended_action": "book_demo",
  "email": "..."
}
```

WEEK 3 — RAG + BUSINESS KNOWLEDGE

Day 15 — Introduction to RAG

RAG means:

Retrieval-Augmented Generation

Problem:

AI does not automatically know private company information.

Solution:

```
Question
↓
Search Company Knowledge
↓
Retrieve Relevant Information
↓
Send Information to GPT
↓
Answer
```

Use Cases

- Company policies
- Product manuals
- Customer FAQs
- Contracts
- Internal documentation
- Training materials

Day 16 — RAG Architecture

Teach:

```
Documents
↓
Extract Text
↓
Split Into Chunks
↓
Create Embeddings
↓
Vector Database
↓
User Question
↓
Similarity Search
↓
Relevant Chunks
↓
GPT
↓
Answer
```

Focus on understanding rather than mathematics.

Day 17 — Embeddings and Vector Search

Explain embeddings simply

Text:

```
How can I return an item?
```

is semantically similar to:

```
Product refund and return policy
```

even though the words differ.

Topics

- Embeddings
- Vectors
- Semantic similarity
- Keyword search vs semantic search
- Vector databases

Introduce:

- pgvector
- Qdrant
- Pinecone
- managed AI retrieval systems

Day 18 — Chunking Documents

Explain why a 200-page PDF cannot simply be pasted into every prompt.

Topics:

- document splitting
- chunk size
- overlap
- headings
- metadata
- semantic chunks

Example:

```
Employee Handbook

Chunk 1:
Vacation Policy
```

```
Chunk 2:
Sick Leave

Chunk 3:
Remote Work

Chunk 4:
Security Policy
```

Exercise

Students manually divide a sample document into useful chunks.

Day 19 — Retrieval Quality

Topics:

- Top-K retrieval
- Relevance
- Metadata filters
- Search quality
- Bad retrieval
- Context overload
- Reranking
- Citations

Exercise

Question:

```
Can employees work internationally?
```

Students decide which retrieved chunks should be passed to GPT.

Day 20 — Building a Knowledge Assistant

Architecture:

```
Company Documents
        ↓
   Knowledge Base
        ↓
```

```
      Search
        ↓
       GPT
        ↓
Answer + Source
```

Exercise

Create the prompt design for an internal company assistant.

Day 21 — RAG Mini Project

Project: Company Knowledge AI

Sample documents:

```
Return Policy
Shipping Policy
Pricing
FAQ
Product Information
```

User:

```
Can I return a product after 20 days?
```

System:

```
Search documents
↓
Find policy
↓
Provide grounded answer
↓
Show source
```

WEEK 4 — n8n + APIs + BUSINESS AUTOMATION

Day 22 — Introduction to APIs

Explain APIs without heavy programming.

Restaurant analogy:

```
Customer
↓
Waiter
↓
Kitchen
↓
Waiter
↓
Customer
```

Software:

```
Application
↓
API Request
↓
Server
↓
API Response
↓
Application
```

Teach:

```
GET
POST
PUT
PATCH
DELETE
```

and:

- JSON
- authentication

- API keys
- bearer tokens

Day 23 — Webhooks and Real-Time Automation

API:

```
You ask another system for information.
```

Webhook:

```
Another system tells you something happened.
```

Example:

```
New Shopify Order
↓
Webhook
↓
n8n
↓
AI
↓
Slack Notification
```

Exercise

Design three webhook automations.

Day 24 — Introduction to n8n

Explain n8n as a workflow automation platform.

Basic workflow:

```
Trigger
↓
Get Data
↓
```

```
AI
↓
Decision
↓
Action
```

Teach:

- Trigger nodes
- Action nodes
- Connections
- Data
- Expressions
- Credentials

Live Demo

```
Manual Trigger
↓
GPT
↓
Output
```

Day 25 — n8n + GPT

Create:

```
Customer Email
↓
n8n
↓
GPT
↓
Classify Email
↓
Route
```

Branches:

```
Sales
Support
```

```
Refund
Spam
```

Teach:

- AI nodes
- IF
- Switch
- expressions
- structured output

Day 26 — n8n + External APIs

Architecture:

```
Webhook
↓
n8n
↓
GPT
↓
HTTP Request
↓
External API
↓
Response
```

Examples:

- Shopify
- CRM
- Slack
- Gmail
- Notion
- Airtable
- Google Sheets

Exercise

Design an API-based order lookup workflow.

Day 27 — Business Automation Patterns

Introduce five important patterns.

Pattern 1

```
Trigger → AI → Action
```

Pattern 2

```
Trigger → AI → Decision → Multiple Actions
```

Pattern 3

```
Trigger → RAG → GPT → Response
```

Pattern 4

```
Trigger → Agent → Tools → Action
```

Pattern 5

```
Trigger → AI → Human Approval → Action
```

Exercise

Students identify which pattern suits five business problems.

Day 28 — Complete Customer Support Automation

Architecture:

```
Email / Website
       ↓
      n8n
       ↓
 AI Classification
       ↓
   AI Agent
```

```
    ├── RAG
    ├── Order API
    ├── CRM
    └── Ticket System
           ↓
     Decision
    ┌────┴────┐
Automatic   Human
 Response   Review
```

Explain each component.

FINAL PROJECT

Day 29 — Design an AI Business Automation System

Students choose one area:

- Customer Support
- Sales
- Marketing
- HR
- Operations
- Ecommerce

Design:

```
Business Problem
↓
Trigger
↓
AI
↓
Knowledge
↓
Agent
↓
API
↓
Automation
↓
Business Result
```

Students create:

1. Business problem
2. Architecture
3. GPT role
4. Agent tools
5. RAG requirements
6. API integrations
7. n8n workflow
8. Human approval points
9. Expected business result

Day 30 — Build & Present Final AI Automation Architecture

Final Project

AI Business Automation Assistant

Example:

```
             Customer
                 ↓
           Website / Email
                 ↓
                n8n
                 ↓
            AI Agent
       ┌─────────┼──────────┐
       ↓         ↓          ↓
      RAG     Order API     CRM
       ↓         ↓          ↓
    Policies   Orders    Customer Data
       └─────────┼──────────┘
                 ↓
             AI Decision
          ┌──────┴──────┐
          ↓             ↓
     Auto Response   Human Review
          ↓
      CRM Update
          ↓
      Analytics
```

Final 60-Minute Session

0–10 min: Course review
10–20 min: Architecture review
20–40 min: Student presentations
40–50 min: Improvement discussion
50–55 min: Career opportunities
55–60 min: Next learning roadmap

Skills Students Have After 30 Days

Students should understand:

GPT

- LLM fundamentals
- Prompt engineering
- Structured outputs
- Business AI use cases

AI Agents

- Agent architecture
- Tools
- Functions
- Memory
- Decisions
- Single vs multi-agent systems

RAG

- Documents
- Embeddings
- Vector search
- Chunking
- Retrieval
- Citations

n8n

- Workflows
- Triggers
- AI nodes
- Conditions
- HTTP requests
- Business automation

APIs

- REST APIs
- JSON
- Authentication
- Webhooks
- API integration

Business Automation

- Customer service
- Sales automation
- Operations
- Marketing
- Human-in-the-loop automation

Recommended Course Weight

Because the course should be **AI-focused**, use approximately:

Subject	Course Weight
GPT / LLM	25%
AI Agents	25%
RAG	20%
n8n	15%
API Integration	10%
Business Automation	5%

This prevents the class from turning into a generic n8n or programming course.

Teaching Principle

Every lesson should answer three questions:

1. What is it?

Explain the concept simply.

2. Why do businesses need it?

Connect the concept to a real problem.

3. How does it work?

Show a practical example.

For example:

```
RAG

WHAT?
A method for giving AI external knowledge.

WHY?
GPT does not automatically know private company documents.

HOW?
Question
→ retrieve documents
→ GPT
→ grounded answer
```

That teaching style works especially well for beginners.

Recommended Final Outcome

After 30 lectures, students should be able to look at a business problem such as:

> "Our support team manually answers hundreds of repetitive questions."

and design:

```
Customer
↓
n8n
↓
GPT Agent
↓
RAG
↓
Customer / Order API
```

```
↓
Automated Response
↓
Human Escalation
```

That should be the central goal of the entire course:

Move students from "How do I use ChatGPT?" to "How do I design an AI-powered business workflow?"